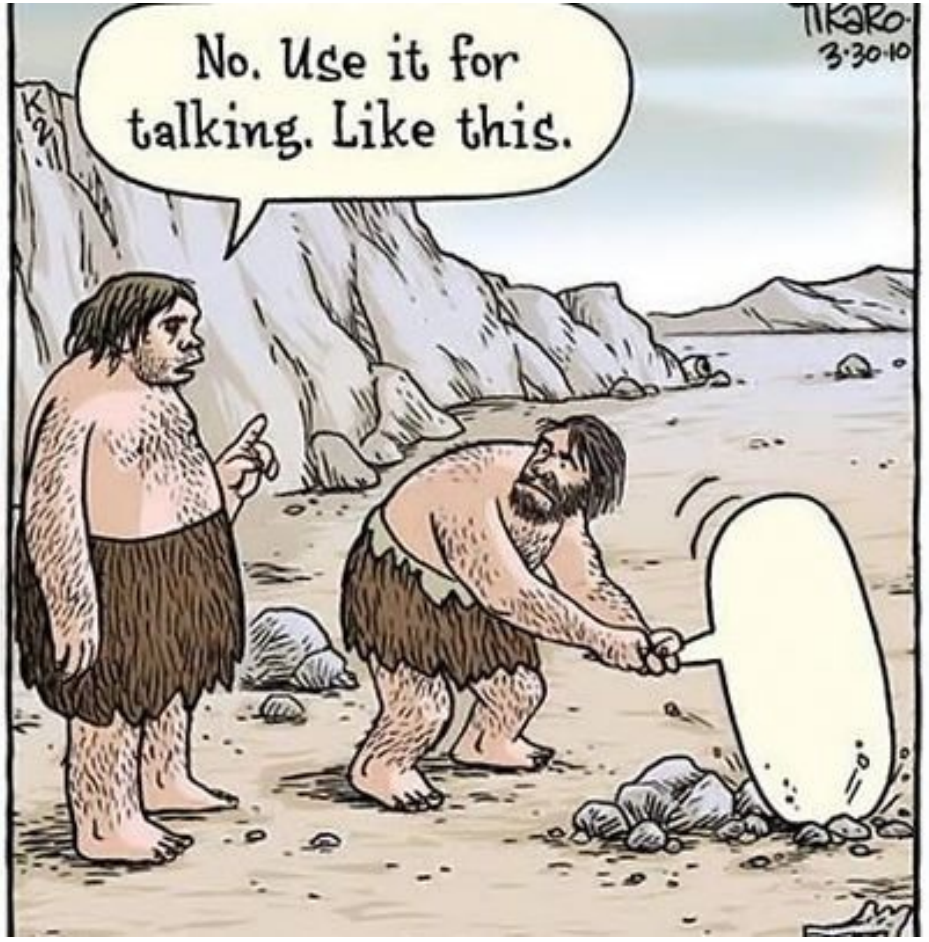
Natural Language Processing

Text Features and when using them



Stylistic features

https://pdfs.semanticscholar.org/d069/8589d7e6cb2f87fb5c4f58189beccb2b526a.pdf

		Languages			
No.	**Feature**	**English**	**Dutch**	**Spanish**	**Italian**
1.	Percentage of Question Sentences	Yes	Yes	Yes	Yes
2.	Percentage of Short Sentences	Yes	Yes	Yes	Yes
3.	Percentage of Long Sentences	Yes	Yes	Yes	Yes
4.	Average Sentence Length	Yes	Yes	Yes	Yes
5.	Average Word Length	Yes	Yes	Yes	Yes
6.	Percentage of Words with Six and More Letters	Yes	Yes	Yes	Yes
7.	Percentage of Words with Two and Three Letters	Yes	Yes	Yes	Yes
8.	Percentage of Semicolons	Yes	Yes	Yes	Yes
9.	Percentage of Punctuations	Yes	Yes	Yes	Yes
10.	Percentage of Pronouns	Yes	----	----	----
11.	Percentage of Prepositions	Yes	----	----	----
12.	Percentage of Coordinating Conjunctions	Yes	----	----	----
13.	Percentage of Comma	Yes	Yes	Yes	Yes
14.	Percentage of Articles	Yes	----	----	----
15.	Percentage of Words with One Syllable	Yes	----	----	----
16.	Percentage of Words with Three Plus Syllables	Yes	----	----	----
17.	Average Syllables per Word	Yes	----	----	----
18.	Percentage of Adjectives	Yes	----	----	----
19.	Percentage of Adverbs	Yes	----	----	----
20.	Percentage of Capitals	Yes	Yes	Yes	Yes
21.	Percentage of Colons	Yes	Yes	Yes	Yes
22.	Percentage of Determiners	Yes	----	----	----
23.	Percentage of Digits.	Yes	Yes	Yes	Yes
24.	Percentage of Full stop	Yes	Yes	Yes	Yes
25.	Percentage of Interjections	Yes	----	----	----
26.	Percentage of Modals	Yes	----	----	----
27.	Percentage of Nouns	Yes	----	----	----
28.	Percentage of Personal Pronouns	Yes	----	----	----
29.	Percentage of Verbs	Yes	----	----	----

Table3. 1: List of all the stylistic features that are used in the development of the author profiling detection system.

Style: n-grams, collocations

Co-occurence of the same n characters or words

Style: POS (Part of Speech)

Playing a particular role (“part”) in a sentence (“speech”) is a feature of one word inside a sentence: no need of full document or corpus.

The number of words having a given POS (number of pronouns, number of adjectives...), possibly normalized w.r.t. the document's length, is a feature of a document.

Style: FW (Function Word)

In linguistics, function words (also called functors) are words that have little lexical meaning or have ambiguous meaning and express grammatical relationships among other words within a sentence, or specify the attitude or mood of the speaker.

Being a FW is a feature of the word itself: if the word is not ambiguous, there is not even need to have the sentence the word belongs to

The number of FW, possibly normalized w.r.t. the document's length, is a feature of a document.

Style: FW (Function Word)

The following is a list of the kind of words considered to be function words:

articles — the and a. In some inflected languages, the articles may take on the case of the declension of the following noun.

pronouns — inflected in English, as he — him, she — her, etc.

conjunctions — and -- uninflected in English

subordinating conjunctions — if, then, well, however, thus, etc.

auxiliary verbs — forming part of the conjugation (pattern of the tenses of main verbs), always inflected

particles -- up, on, down

interjections — sometimes called "filled pauses"

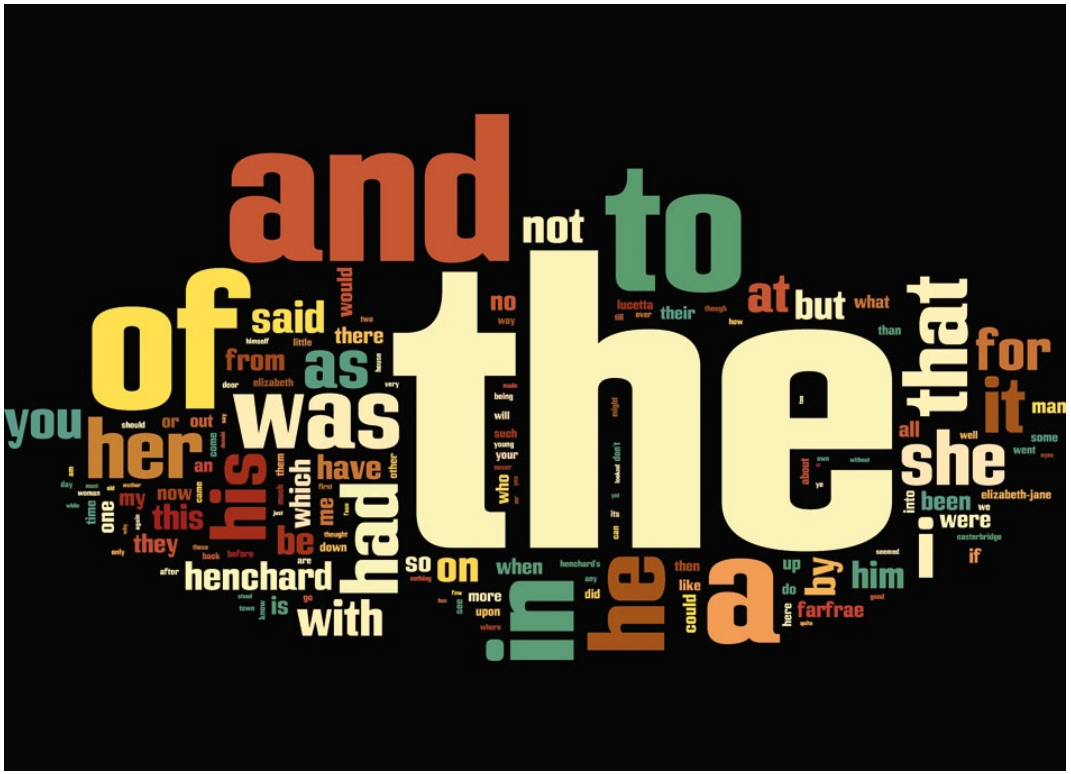
expletives — take the place of sentences, among other functions.

pro-sentences — yes, okay, etc

https://en.wikipedia.org/wiki/Function_word

Style: FW (Function Word)

Huge overlap with stop words



Style: readability measure

Readability is a measure of how easy a piece of text is to read. It can include elements of complexity, familiarity, legibility and typography. Readability formulas usually look at factors like sentence length, syllable density and word familiarity as part of their calculations.

https://readable.com/readability/

https://app.readable.com/text/

Content-based features

Frequency of words belonging to some given, knows semantic category (dictionary-based, topic-based)

Emotion words

TF-IDF (recall:

tf-idf(t, d, C) = tf(t, d) * idf(t, C) where

tf(t,d) = number of t in d / length(d) and

idf(t, C) = log (|C|/(|documents in C which contain t|))

)

LSA (Latent Semantic Analysis, implemented by word2vect)

Content- & style-based features

LIWC (Linguistic Inquiry and Word Count): http://liwc.wpengine.com/

Most of the execution time spent to answer these questions is dedicated to writing the output files: the reasoners produce a written output for successive inspection. Except for Q2, by increasing the size of the log file to analyse, the time required to answer the questions suffers from a minor increase. Answering Q2 heavily depends on the log file size because DigForReas-J has to check combinations of multiple agents. The other questions instead can be reduced to a straightforward reading of the log file, with time complexity linear in the log file length.

Content- & style-based features

LIWC (Linguistic Inquiry and Word Count): http://liwc.wpengine.com/

TRADITIONAL LIWC DIMENSION	YOUR DATA	AVERAGE FOR PROFESSIONAL OR SCIENTIFIC WRITING
I-WORDS (I, ME, MY)	0.0	0.63
SOCIAL WORDS	0.0	7.62
POSITIVE EMOTIONS	0.0	2.32
NEGATIVE EMOTIONS	1.1	1.45
COGNITIVE PROCESSES	17.0	7.52
SUMMARY VARIABLES		
ANALYTIC	99.0	92.57
CLOUT	37.5	68.17
AUTHENTICITY	65.6	24.84
EMOTIONAL TONE	11.9	43.61

Vocabulary & Idiosyncrasies

Unique words: Number of unique words in the sample

Capital words: Number of words (length > 2) containing only capital letters

Spelling errors: Number of words that are not found in the language dictionaries

Character flooding: sooooo saaaad...

Neologisms

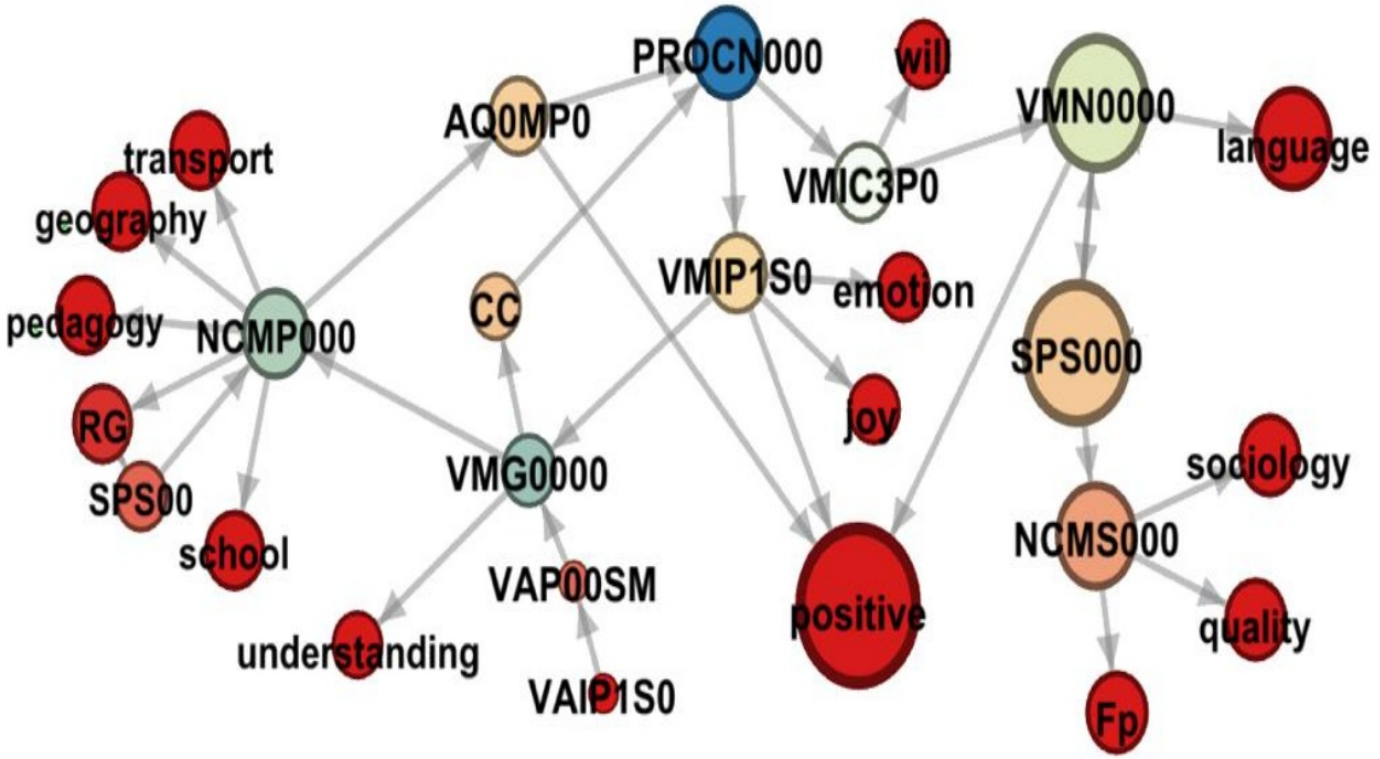
Slang: tanta roba!

Emoticons: >(

Graph-based features

He estado tomando cursos en línea sobre temas valiosos que disfruto estudiando y que podrían ayudarme a hablar en público.

(I) have been taking online courses about valuable subjects that (I) enjoy studying and that might help me to speak in public.



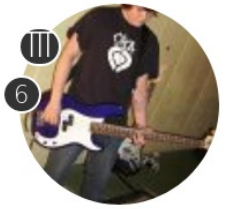
Content, FW, POS: when, why?

Content, FW, POS: when, why?

Well... it depends on what is your goal...

Content, FW, POS: when, why?

If you need to classify a document according to the “music”, “biology”, “food” categories (just three examples of categories taken from BabelNet), maybe you do not care of FW and POS



bass, basso

An adult male singer with the lowest voice

ID: 00008913n | Concept



sea bass, bass

The lean flesh of a saltwater fish of the family Serranidae

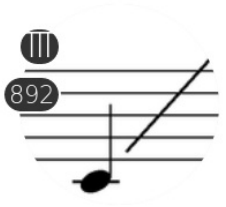
ID: 00008914n | Concept



bass, freshwater bass

Any of various North American freshwater fish with lean flesh (especially of the genus Micropterus)

ID: 00008915n | Concept



bass, bass voice, basso

The lowest adult male singing voice

ID: 00008916n | Concept

Content, FW, POS: when, why?

If you need to classify a document according to the “music”, “biology”, “food” categories (just three examples of categories taken from BabelNet), maybe you do not care of FW and POS

Baked sea bass with lemon caper dressing. To make the dressing, mix the oil with the lemon zest and juice, capers, mustard, some seasoning and 1 tbsp water. Don't add the parsley yet (unless serving straight away) as the acid in the lemon will fade the colour if they are left together for too long.

https://www.bbcgoodfood.com/recipes/baked-sea-bass-lemon-caper-dressing

Content, FW, POS: when, why?

If you need to classify a document according to the “music”, “biology”, “food” categories (just three examples of categories taken from BabelNet), maybe you do not care of FW and POS

```
These Russian basses sing so low you’ll think
the world is ending. What’s that low rumble?
Is the sky falling in? Is it a plane? Or is
it the sound of a Russian oktavist, singing a
full octave below a normal bass…?
```

```
https://www.classicfm.com/music-
news/videos/oktavist-low-bass/
```

Content, FW, POS: when, why?

But if you are profiling the author of a document...

Content, FW, POS: when, why?

But if you are profiling the author of a document...

"However, the preponderance of evidence suggests that ***the most consistently effective features over a wide variety of authorship problems are function words and individual parts-of-speech****"*

Shlomo Argamon, Moshe Koppel, James W. Pennebaker, and Jonathan Schler. 2009. Automatically profiling the author of an anonymous text. Commun. ACM 52, 2 (February 2009), 119-123. DOI: https://doi.org/10.1145/1461928.1461959

http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.136.9952&rep=rep1&type=pdf

Content, FW, POS: when, why?

Female vs. male

My aim in this article is to show that given a relevance theoretic approach to utterance interpretation, it is possible to develop a better understanding of what some of these so-called apposition markers indicate. It will be argued that the decision to put something in other words is essentially a decision about style, a point which is, perhaps, anticipated by Burton-Roberts when **he** describes loose apposition as a rhetorical device. However, **he** does not justify this suggestion by giving the criteria for classifying a mode of expression as a rhetorical device. Nor does **he** specify what kind of effects might be achieved by a reformulation or explain how it achieves those effects. In this paper **I** follow Sperber and Wilson's (1986) suggestion that rhetorical devices like metaphor, irony and repetition are particular means of achieving relevance. As **I** have suggested, the corrections that are made in unplanned discourse are also made in the pursuit of optimal relevance. However, these are made because the speaker recognises that the original formulation did not achieve optimal relevance .

The main aim of this article is to propose an exercise in stylistic analysis which can be employed in the teaching of English language. It details the design and results of a workshop activity on narrative carried out with undergraduates in a university department of English. The methods proposed are intended to enable students to obtain insights into aspects of cohesion and narrative structure: insights, it is suggested, which are not as readily obtainable through more traditional techniques of stylistic analysis. The text chosen for analysis is a short story by Ernest Hemingway comprising only 11 sentences. A jumbled version of this story is presented to students who are asked to assemble a cohesive and well formed version of the story. **Their** re-constructions are then compared with the original Hemingway version.

Content, FW, POS: when, why?

Female vs. male

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So, what?

Possibly endless amount of features: many are standard and well-studied ones, and it is also clear which NLP task they are more suitable for.

Others (EmoGraph, as an example) are less standard.

Anyone is able to use existing features, already implemented by existing tools

The real challenge would is to invent new features, or well-calibrated combinations of existing ones, and **demonstrate that they are useful for some given NLP task**, on some existing benchmark.